AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 36 · STUDENT LESSON

ML Builder Expo

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Show the evidence. Explain the failure. Bound the claim.

Defend an auditable project with readable results,honest unknowns and a justified next investigation.

Keep one habit: Complete paper card → pictured indoor-photo limits → executable ledger audit → peer questions → independent95%claim.

MAIN LESSON · SECTION 1

Can someone else check your project story?

> Show the question,method,evidence,failures and limits together.

Defense goals

I can defend the project using traceable **evidence**.

I can explain errors and limits to a non-specialist audience.

I can respond to critique and propose an **evidence**-based next step.

Recall selection,documentation and controlled repair

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Without opening the old lesson, explain one key idea from Capstone Train, Analyze, Improve. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Model Cards and Human Review. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Improve Generalization. Give an example and a mistake that the idea helps you avoid.
- What is the difference between a result and a claim?

End goal

By the end, I can explain ml builder expo using a traced example and a stated **limitation**.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

A builder defends the complete **evidence** chain: problem, data, code, model, metrics, errors, fairness, **limitation**s, and human oversight.



LEARNING TARGET Final counts are TP=12,FN=3,FP=5,TN=20. Compute accuracy, precision and recall, then state one **limitation** and a review trigger for the model **defense**.

Builder name and experiment date:

Readiness for the connected explanation

This week’s end goal

Tell an evidence-linked story in everyday language

> A small or unsuccessful project can still demonstrate strong reasoning.

What a defense does

A defense is an explanation that another person can question and check. Start with the task, point to the saved evidence, explain one failure and state what remains unknown. “I do not have that record yet”is a useful honest answer followed by a concrete request. The example below is a supplied paper record, not a trained system.

Your own project should use its actual saved artifacts; do not copy the paper scores or the parcel kit’s teacher-only final answers into a plant-project claim.

The final **defense** tells a coherent story: question, data, method, comparison, errors and limits. A strong project may conclude that the baseline is preferable or **evidence** is insufficient.

The aim is sound reasoning and a reproducible investigation, not an impressive headline or polished unsupported claim.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines machine learning project **defense**.

▼ Earlier wording, alternatives and picture transcripts

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



MYSTERY The team must use ml builder expo to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

MAIN LESSON · SECTION 3

Inspect a complete bounded paper defense

> The ledger checks arithmetic;it cannot create absent training artifacts.

Complete12-case saved predictions

Paper test ID	Reference	Frozen candidate	Fixed clear baseline
R01	clear	clear	clear
R02	clear	clear	clear
R03	clear	clear	clear
R04	clear	clear	clear
R05	clear	clear	clear
R06	clear	clear	clear
R07	clear	clear	clear
R08	clear	blocked	clear
R09	blocked	clear	clear
R10	blocked	clear	clear
R11	blocked	blocked	clear
R12	blocked	blocked	clear

Complete evidence card,failures and unknowns

Complete paper defense field	Evidence / explicit status
Task and version	Fictional parcel clear/blocked comparisonV1; blocked positive; no real handling decision.
Supplied test premise	12 source-separated test cases under a frozen-procedure premise; exact training/split/code artifacts are not supplied.
Candidate arithmetic	TN 7, FP 1, FN 2, TP 2; 9/12=75%accuracy, 2/3 precision, 2/4 recall.
Same-case baseline	Fixed-clear 8/12≈66.7%; all 4 blocked cases missed. Actual training-majority counts are not supplied for this separate paper fixture.
Complete error records	R08 clear→blocked:false alarm; R09/R10 blocked→clear:misses. Lighting/causes are not supplied.
Readiness and next request	Paper arithmetic can be checked; actual reproducible model/split provenance and human review status remain unprovided. Request these records before a release claim.
Next investigation	If dim-condition coverage is a concern, collect labelled development evidence under a declared protocol; these rows do not identify dim-light failures.

Example opening and its proposed next investigation

Example opening: We compared a simple classifier with an always-clear **training** baseline on fictional parcels. We froze the procedure before a 12-case test. Candidate accuracy was 9/12 versus baseline 8/12; candidate FP 1 FN 2.

These small synthetic results support a classroom comparison, not real parcel decisions. We would next investigate dim-condition coverage using new development data.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

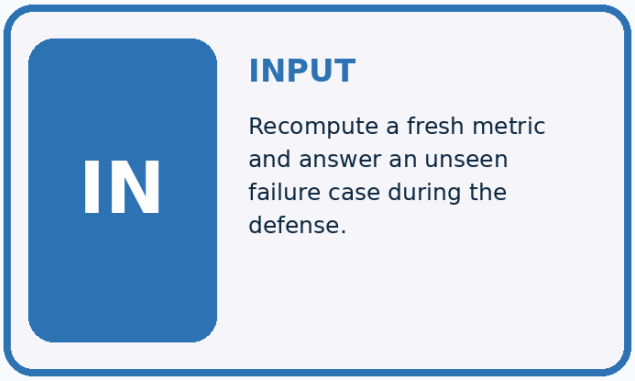
Keep the distinction between an observed result and a proposed next step.

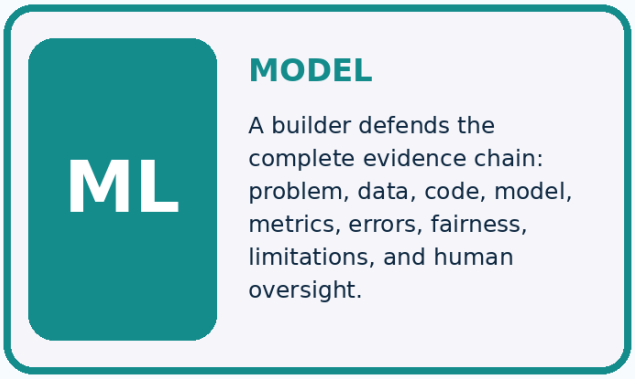
▼ Earlier wording, alternatives and picture transcripts

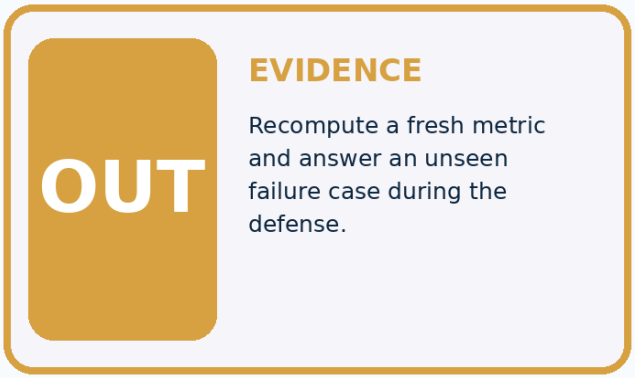
ML **MECHANISM**

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







QUESTION Recompute a fresh metric and answer an unseen failure case during the **defense**.

MODEL IDEA A builder defends the complete **evidence** chain: problem, data, code, model, metrics, errors, fairness, **limitation**s, and human oversight.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated **limitation**.

UNDER THE HOOD Trace one leaf from source and split through pre**process**ing, model, **prediction**, and review. Recompute precision or recall from a fresh confusion matrix without notes.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Connect the picture and the explanation

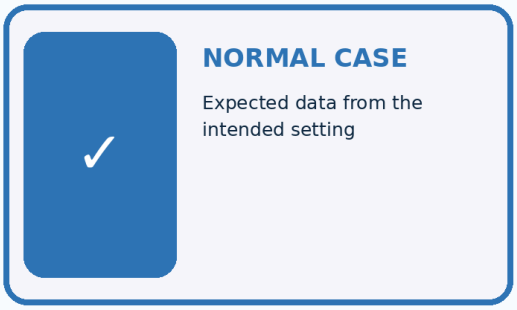
Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

MAIN LESSON · SECTION 4

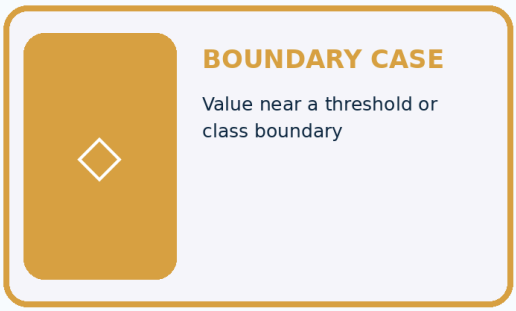
Compare a checkable defense with a polished unsupported claim

> Good presentation makes the evidence readable;it does not add observations.

Four evaluation conditions



NORMAL Expected data and expected behavior.

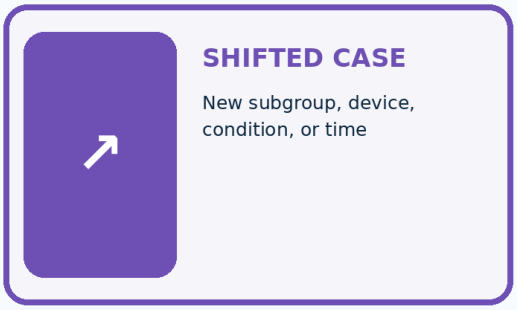


Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.



MISSING Necessary field, label, source, or measurement is unavailable.



SHIFTED New lighting, species, device, subgroup, or time condition.

Keep the claim bounded

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Choose a revealing case

- Which case most threatens today's claim, and why?

Predict the supplied changed case

- Change: The poster omits the baseline and calls the model “field ready”.

▼ **Check after predicting**

Prediction: The headline exceeds the supplied **evidence**.

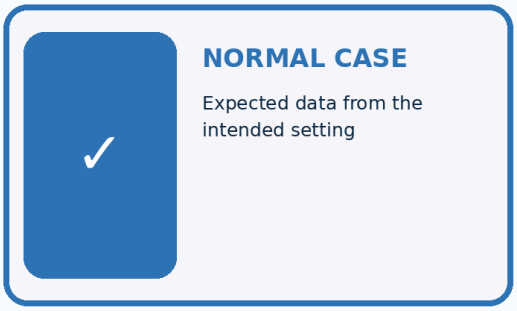
Reason: No outdoor evaluation is supplied, and the comparative benefit is obscured without the same-case baseline.

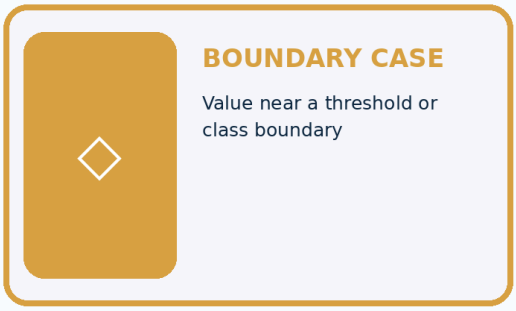
▼ Earlier wording, alternatives and picture transcripts

EVALUATION DESIGN

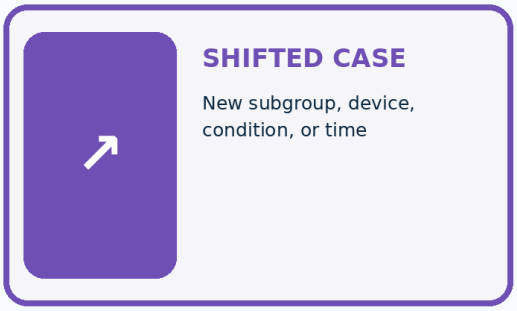
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Put the failure record beside the expo headline

> Five plants,15 related views and indoor-only coverage belong in the claim.

Pictured facts and distinct scope

Present a plant-classifier expo claim that another team could check.

Toy expo **evidence**: 15 held-out photos from 5 plants; candidate12 correct, baseline10 correct. Three candidate failures are recorded; no outdoor trial exists.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture is a separate indoor plant-photo paper fixture:5 plant identities with 3 views each give 15 photo cases. Candidate12/15=80%, baseline 10/15≈66.7%. Related views do not become 15 independent plants. The three candidate failures are represented in the complete

supplementary ledger below; their causes are not established. No outdoor observations are supplied.

Keep this record separate from the main 12-parcel defense and from the 72-row capstone kit.

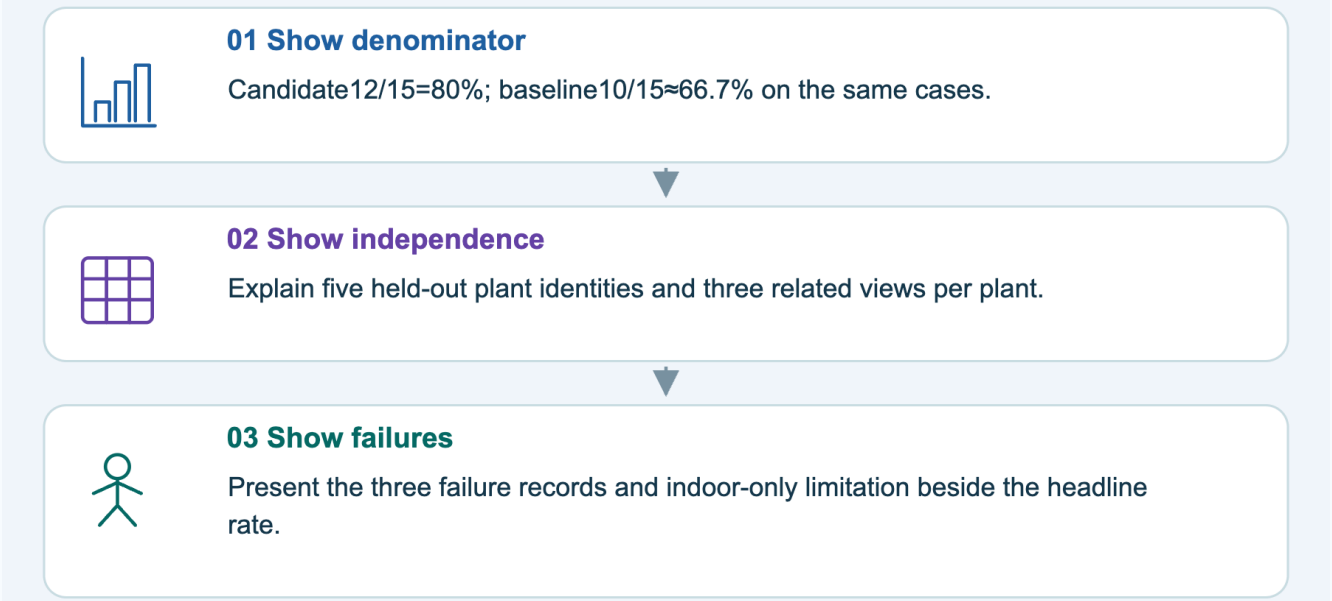
Complete supporting photo/plant ledger

Indoor picture photo / plant group	Toy reference	Candidate	Fixed clear baseline
F01 / G1	clear	clear	clear
F02 / G1	clear	clear	clear
F03 / G1	clear	blocked	clear
F04 / G2	clear	clear	clear
F05 / G2	clear	clear	clear
F06 / G2	clear	clear	clear
F07 / G3	clear	clear	clear
F08 / G3	clear	clear	clear
F09 / G3	clear	clear	clear
F10 / G4	clear	clear	clear
F11 / G4	blocked	clear	clear
F12 / G4	blocked	blocked	clear
F13 / G5	blocked	blocked	clear
F14 / G5	blocked	clear	clear
F15 / G5	blocked	blocked	clear

Trace evidence,failures and limits

AI ACADEMY / AI-4/36 · 02 · TRACE THE CASE

Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Show denominator	Candidate12/15=80%; baseline10/15≈66.7% on the same cases.
2. Show independence	Explain five held-out plant identities and three related views per plant.
3. Show failures	Present the three failure records and indoor-only limitation beside the headline rate.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

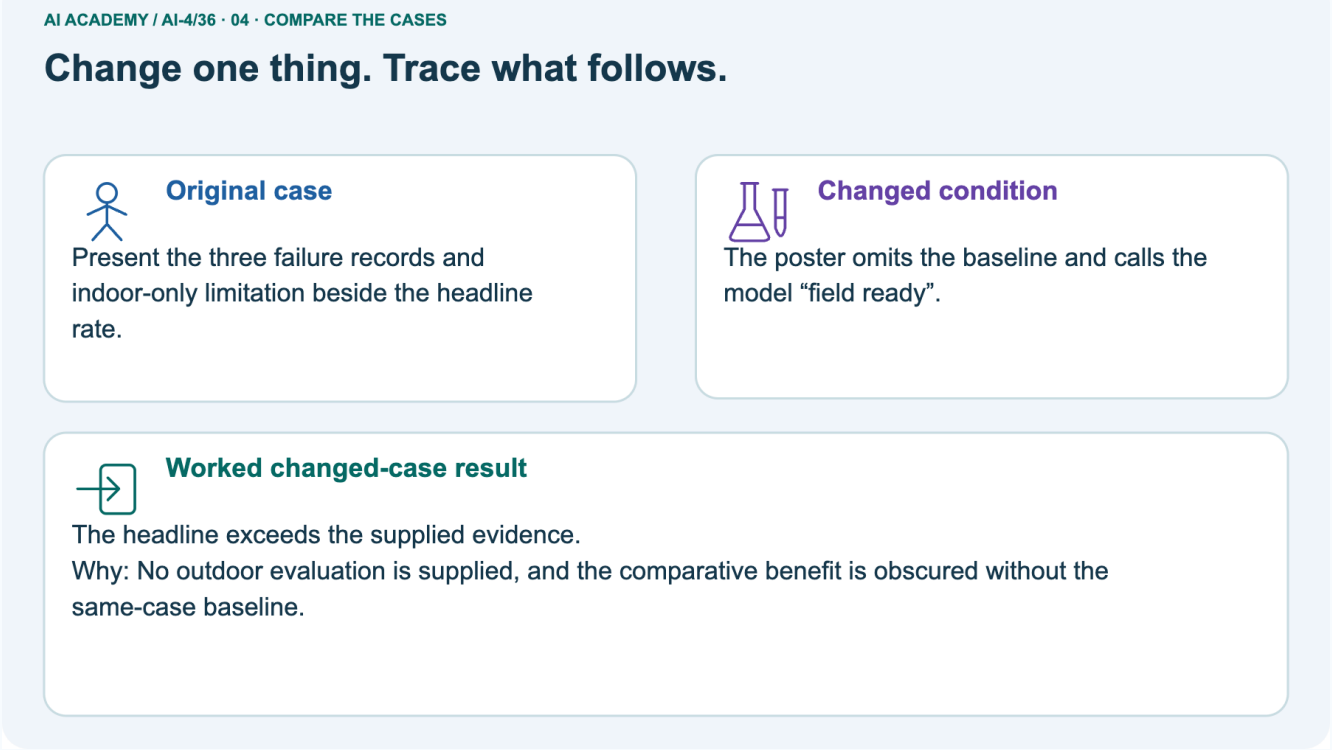
Read the picture in order. Present the three failure records and indoor-only **limitation** beside the headline rate. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict the field-ready publicity claim

- Change one thing: The poster omits the baseline and calls the model “field ready”. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain why polish is insufficient

- Clear presentation helps review but does not increase the strength of a tiny or unrepresentative evaluation.

- Read the labelled picture: Which three supporting items should be visible at the expo? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Return to the evidence-file audit

Machine learning project defense at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Show the process	Task data splits versions	Explain one prediction
Show the evidence	Baseline metrics errors	Trace one reported number
Show the limits	Unsupported uses next step	Answer peer questions

Prepare a five-minute presentation with a one-page evidence summary and linked project files. Include the data dictionary, split record, run record, prediction ledger, model card and revision history. Demonstrate one prediction or arithmetic calculation live so the audience can follow the reasoning.

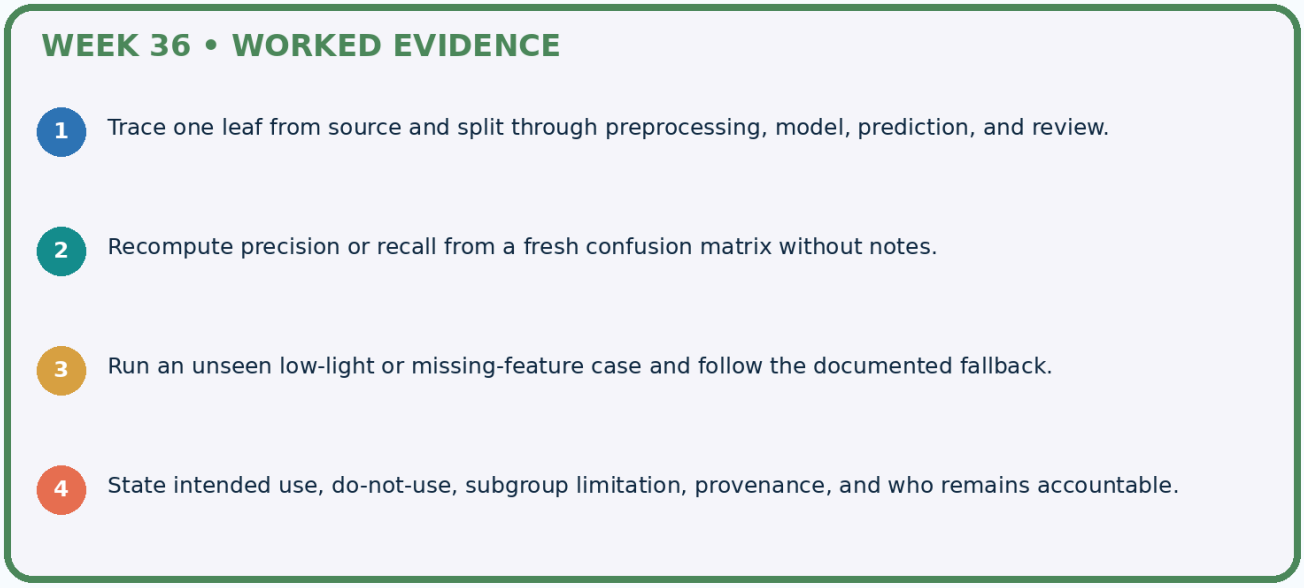
Name the files needed to audit one reported accuracy figure. Explain the steps or evidence behind your response.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.



STEP 1 Trace one leaf from source and split through pre**process**ing, model, **prediction**, and review.

STEP 2 Recompute precision or recall from a fresh confusion matrix without notes.

STEP 3 Run an unseen low-light or missing-feature case and follow the documented fallback.

STEP 4 State intended use, do-not-use, subgroup **limitation**, provenance, and who remains accountable.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: ML Builder Expo

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find the first unsupported step in a smooth defense

> A serious data problem cannot be hidden by presentation marks.

Challenge the demonstration-only claim

- Diagnose one cause and name the cleanest next experiment:

CLAIM TO CHALLENGE A smooth prepared demo proves the model is reliable and ready for use.

Read the four separate rubric dimensions

The teacher rubric has four dimensions scored 0, 1, 2: valid data/splits, reproducible method, complete evaluation, and honest communication.

Two means **evidence** is complete and explainable; one means partial **evidence** with identified gaps; zero means absent or materially incorrect.

Report dimensions separately; a serious leakage problem cannot be hidden by presentation points.

Choose the next investigation

> **Write one next investigation tied to an observed limitation, and identify what must stay held out.**

▼ Earlier wording, alternatives and picture transcripts

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: A smooth prepared demo proves the model is reliable and ready for use. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

A closer explanation

MAIN LESSON · SECTION 8

Answer peer questions with records and honest unknowns

> Mark feedback resolved,deferred or requiring new evidence.

Try the supported evidence-file audit

Use workbook W2 for the connected example “Machine learning project **defense**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Ask and record the three review questions

Peers ask three questions: what influenced model selection, which cases failed, and what **evidence** would change the conclusion? Respond using saved records. If an answer is unknown, say so and name the **evidence** needed.

Record feedback as resolved, deferred with reason, or requiring new data.

Explain the evidence beyond a final answer

- Use the worked case for Machine learning project **defense**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Answer the pictured supporting-items question

- Which three supporting items should be visible at the expo? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

Reserve the fresh95%claim

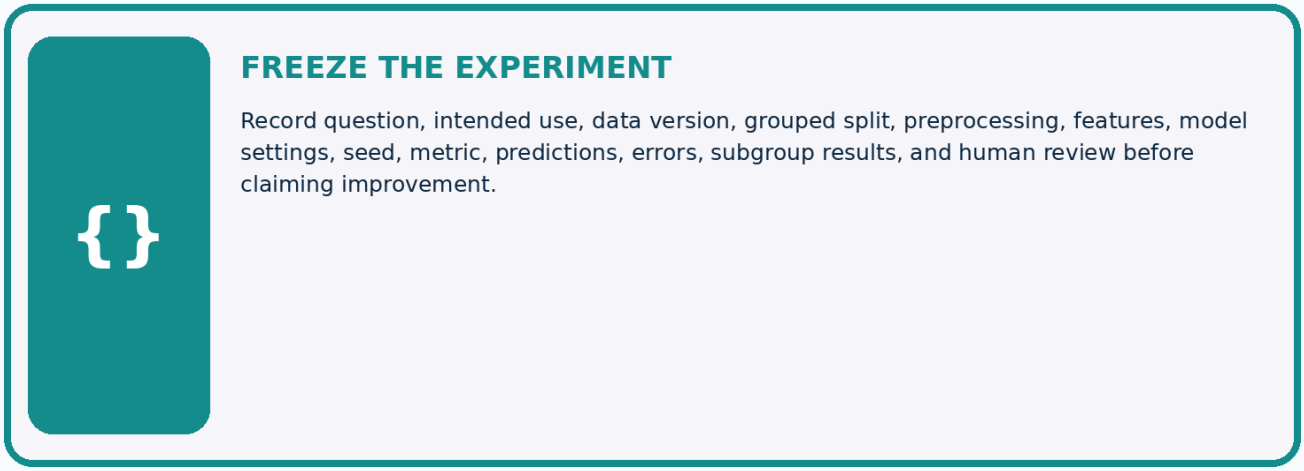
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + **limitation**s

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Practise with less help: Trace evidence

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run a complete evidence-audit and peer-review expo

> Every shared question has supplied inputs and an honest answer boundary.

Complete runnable paper arithmetic audit

```
# Complete fictional paper expo ledger; this is not a fitted classifier.
records = [
    {'id': f'R{i:02}', 'reference': 'clear' if i <= 8 else 'blocked',
     'candidate': 'clear' if i <= 7 or 9 <= i <= 10 else 'blocked',
     'baseline': 'clear'} for i in range(1, 13)
]
def audit(rows, field):
    if not rows or len({r['id'] for r in rows}) != len(rows):
        raise ValueError('Need nonempty uniquely identified records')
    cells = [0, 0, 0, 0]  # TN, FP, FN, TP; blocked positive
    for r in rows:
        if r['reference'] not in ('clear', 'blocked') or r[field] not in ('clear',
'blocked'):
            raise ValueError('Missing reference or prediction; do not guess')
        cells[2 * (r['reference'] == 'blocked') + (r[field] == 'blocked')] += 1
    tn, fp, fn, tp = cells
    return {'cells': cells, 'correct': tn + tp, 'total': len(rows),
            'precision': tp / (tp + fp) if tp + fp else None,
```

```
            'recall': tp / (tp + fn) if tp + fn else None,
            'errors': [r['id'] for r in rows if r[field] != r['reference']]}
print('candidate', audit(records, 'candidate'))
print('baseline', audit(records, 'baseline'))
print('Paper arithmetic reproduced; actual training/code/split provenance remains
unprovided.')
```

Five complete reviewer questions

Complete reviewer question · Evidence-linked answer / next action

> **Complete reviewer question:** What influenced selection?
>
> **Evidence-linked answer / next action:** The paper premise says frozen before test, but does not provide exact selection logs. Request the original data/code/split/run/freeze records; do not claim you have inspected them.

> **Complete reviewer question:** Where did it fail?
>
> **Evidence-linked answer / next action:** R08 false alarm, R09/R10 misses; no lighting/source-condition cause is given. Show all three and request condition records if needed.

> **Complete reviewer question:** Is it better than baseline?
>
> **Evidence-linked answer / next action:** 9/12 versus 8/12 on these cases; one false alarm added, two blocked misses prevented. No universal gain is established.

> **Complete reviewer question:** What would change your conclusion?
>
> **Evidence-linked answer / next action:** Invalid split/provenance, unreviewed labels or unsupported intended-use coverage would require investigation and a revised claim. Collect new development evidence, then new frozen final evidence for a changed version.

> **Complete reviewer question:** Can you release it for real use?

> **Evidence-linked answer / next action:** This supplied fictional paper record supports a classroom audit only. No runnable classifier artifacts, actual teacher approval or real-use readiness is established.

Normal,boundary,missing and shifted situations

Complete expo probe	Defensible action
Normal complete twelve-pair ledger	Recompute candidate 9/12, baseline 8/12; precision 2/3, recall 2/4.
Boundary zero predicted positives in clear baseline	Precision undefined, not perfect; recall 0/4.
Missing model/code/split artifacts	Say not supplied; request them. A ledger alone cannot reproduce training.
Missing plant feature at a live demo	Follow the project's documented reject/review policy. If none exists, stop and document that gap; do not fabricate a prediction.
Shifted outdoor plant request	No outdoor coverage is supplied in the pictured record; mark unsupported and propose a separate evidence collection rather than claiming the indoor score applies.

Present,review,swap and retain responses

1. Expo Defense:in pairs, one presenter uses the complete paper card, ledger and program while the reviewer asks the five supplied questions. Swap roles and retain each learner's explanation and feedback status. For your actual project, show the real saved artifacts and actual results instead of this paper fixture.

   The normal, boundary, missing and shifted probes above are complete audit situations, not claimed runs of an absent classifier. Keep the selected 95%claim for an uncoached first review; the teacher may assign a separate unseen calculation after checking its key.

Keep the evidence ledger

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

> **Case:** Boundary
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

> **Case:** Missing
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

> **Case:** Shifted
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

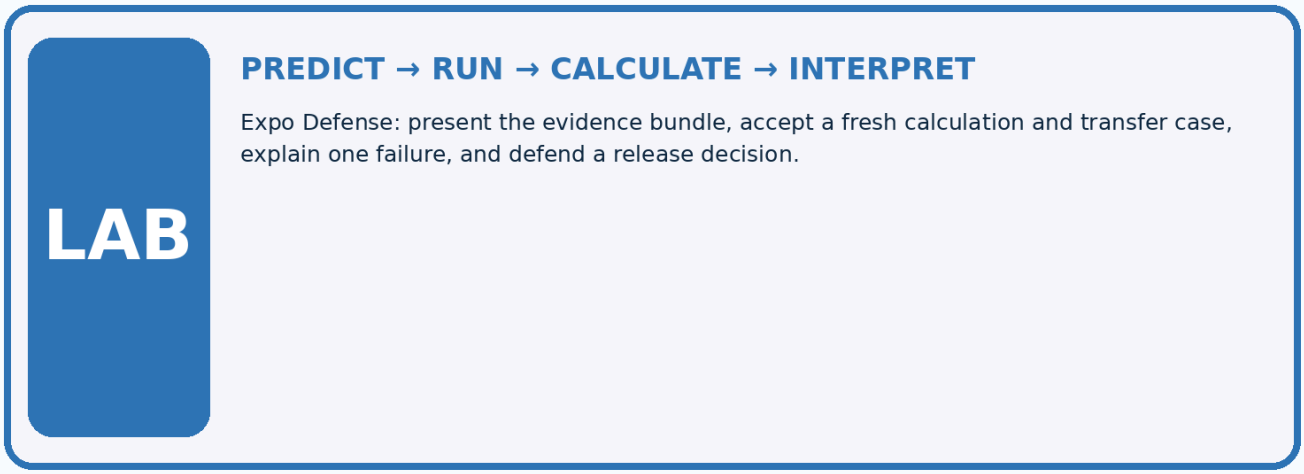
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.



1. Write the hypothesis and expected **evidence** before running.

2. Expo **Defense**: present the **evidence** bundle, accept a fresh calculation and transfer case, explain one failure, and defend a **release decision**.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: review a fresh polished95%claim

> Give three concrete evidence requests and explain the remaining use limit.

Complete selected defense claim and scope

Review a **defense** claiming 95 percent success on 20 cases but omitting the baseline, split history and errors. Give three revision requests and explain why polish is insufficient.

The selected independent defense reports 95%on20 cases but omits its baseline, split history and errors. Treat95%as a reported claim:if exact and all 20 scored, it would mean 19 correct, but no ledger verifies that. Request the missing evidence rather than invent a confusion matrix or assume reliability across schools. Keep this case separate from the worked 12-case defense and pictured 15-photo record.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Apply independently

Week 36 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Your expo model succeeds on three demonstration plants. What **evidence** should accompany the demo and what **limitation** should you state?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Deliver the final Responsible Plant Classifier product card, live **evidence** trace, model card, and human-oversight commitment.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

Final counts are TP=12,FN=3,FP=5,TN=20. Compute accuracy, precision and recall, then state one **limitation** and a review trigger for the model **defense**.

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Leave one complete project evidence bundle

> The project's own files and limits should support every reported result.

Final artifact and evidence

THIS WEEK'S CHECKPOINT Deliver the final Responsible Plant Classifier product card, live **evidence** trace, model card, and human-oversight commitment.

Deliver the actual project question, label definition, data/source/split record, code/preprocessing/model versions, selection/freeze history, baseline and candidate prediction ledgers, metrics/errors/slices, model card and actual review status. Trace one case from available input through the saved model to evaluation and fallback. State which checks were run, traced, not run or unsupported.

A polished poster and copied example scores do not replace these artifacts. Propose one next investigation tied to a documented limitation, and reserve fresh final evidence for a later changed version.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and **limitation** of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Versions,uncertainty and human review

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



Evidence for your project

MAIN LESSON · SECTION 16

Finish with a claim you can defend and a limit you can explain

> Use the evidence to choose the next learning step.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Defend the project using traceable **evidence**

My deciding step, evidence and one remaining question

Plan an uncoached changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the current experiment**

REFERENCE · SECTION 5

Reference · Words for the current experiment

> Tie each word to a particular input, fitted choice or evidence record.

Core vocabulary

Term	Precise meaning	My ML example
technical defense	evidence-based explanation under questions	
traceability	link claim to data, code, result, and decision	
transfer test	fresh case not rehearsed	
release decision	approve, limit, revise, or stop	

Closer-explanation terms

Word	Meaning
Defense	An explanation of decisions supported by inspectable evidence
Traceability	Ability to follow a claim back to its source
Limitation	A boundary on interpretation or supported use

Word	Meaning
Reflection	An account of learning and justified next steps

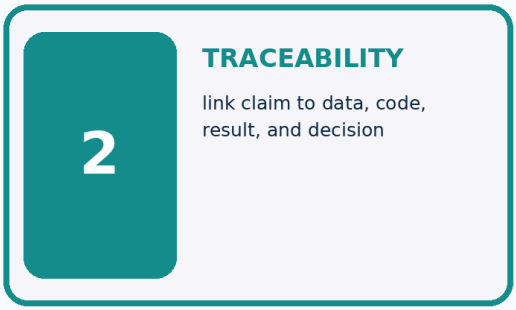
▼ Earlier wording, alternatives and picture transcripts

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









MISCONCEPTION TO REPAIR A smooth prepared demo proves the model is reliable and ready for use.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

▼ **Extra challenge · Build a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Build a stronger explanation

> Choose the next useful challenge after the main route.

Thinking ladder

- DEFINE Use all four terms precisely: **technical defense**, **traceability**, **transfer test**, **release decision**.
- CALCULATE Reproduce the worked **evidence** with units and denominator.
- TRACE Explain the data, code or decision path.
- DIAGNOSE A smooth prepared demo proves the model is reliable and ready for use.
- TRANSFER Solve a fresh dataset or model case.
- IMPROVE Change one cause and rerun fixed **evidence**.
- DEFEND State intended use, metric, **limitation**, and human responsibility.

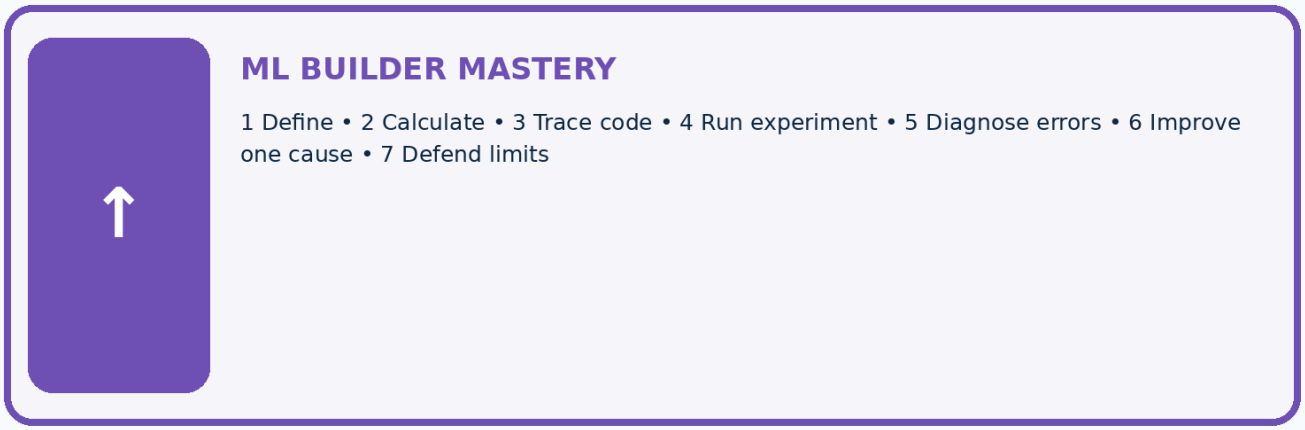
MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



▼ Reference · Inspect your evidence

REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use after practice; keep the selected fresh95%defense claim for the individual attempt.

Experiment evidence checklist

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a **limitation**, fairness concern, and the human-review action.
- Independent **evidence** and **defense**:

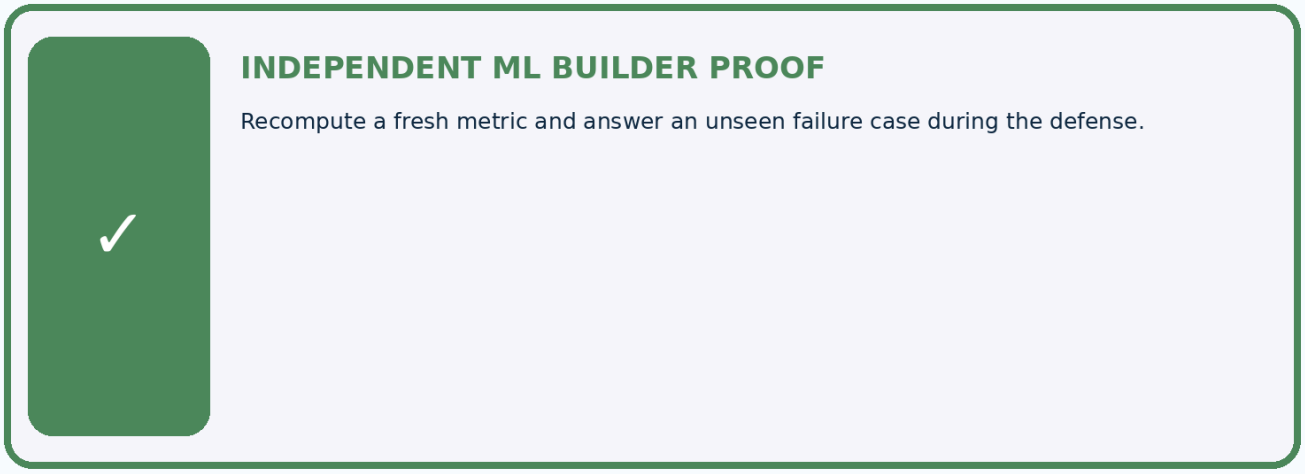
- My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



REQUIRED PROOF Recompute a fresh metric and answer an unseen failure case during the **defense**.

NEXT BRIDGE Carry the experiment **evidence** into: AI-5 AI Programmer.

▼ Extra practice · Earlier complete case

Extra practice · Earlier complete case

> Attempt the question before opening its worked explanation.

Situation and question

Final classifier gets16/20 correct; two of four errors are false negatives.

Report supported metrics and what is missing.

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

Accuracy**80%**. Recall cannot be computed from FN alone; actual positives or TP are needed.
Show full matrix, limits and review path.

Evidence or stage	Value or result
Correct final predictions	16/20

▼ Earlier wording, alternatives and picture transcripts

Week 36 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Report only metrics the evidence supports

Final classifier gets16/20 correct; two of four errors are false negatives.

Report supported metrics and what is missing.

Accuracy**80%**. Recall cannot be computed from FN alone; actual positives or TP are needed. Show full matrix, limits and review path.

▼ **Reference · Earlier route checklist**

REFERENCE · SECTION 15

Reference · Earlier route checklist

> Use one main route; revisit the checklist only when useful.

Checklist and first prediction

By the end, I can	How I will show it
Defend the project using traceable evidence	Explain the deciding step, show a trace or test, and state one limit.
Explain errors and limits to a non-specialist audience	Explain the deciding step, show a trace or test, and state one limit.
Respond to critique and propose an evidence-based next step	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Earlier wording, alternatives and picture transcripts**

Start with this question: Present a plant-classifier expo claim that another team could check.

▼ **Extra practice · Predict, check and repair**

REFERENCE · SECTION 17

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Present a plant-classifier expo claim that another team could check.
- Which labelled step matches this detail? Present the three failure records and indoor-only **limitation** beside the headline rate.
- 1. Show failures
- 2. Show independence
- 3. Show denominator
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Show failures Present the three failure records and indoor-only **limitation** beside the headline rate. Read the picture in order.

Present the three failure records and indoor-only **limitation** beside the headline rate. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: The poster omits the baseline and calls the model “field ready”. Which conclusion is supported by the supplied facts?
- 1. The headline exceeds the supplied **evidence**.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: The headline exceeds the supplied **evidence**. No outdoor evaluation is supplied, and the comparative benefit is obscured without the same-case baseline.

Repair and keep the record

Review Capstone Train, Analyze, Improve (ai-4/35). Goal: Run a frozen final evaluation with complete accounting. Short practice: Why is “the revision is better” incomplete?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Model Cards and Human Review (ai-4/33). Goal: Write a model card whose claims match **evidence**. Short practice: Write a bounded replacement for the publicity sentence.

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Improve Generalization (ai-4/30). Goal: Plan one **evidence**-led change and test it on source-separated validation cases. Short practice: What fresh **evidence** is needed after adding crossed **training** data? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, **reflection**s, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

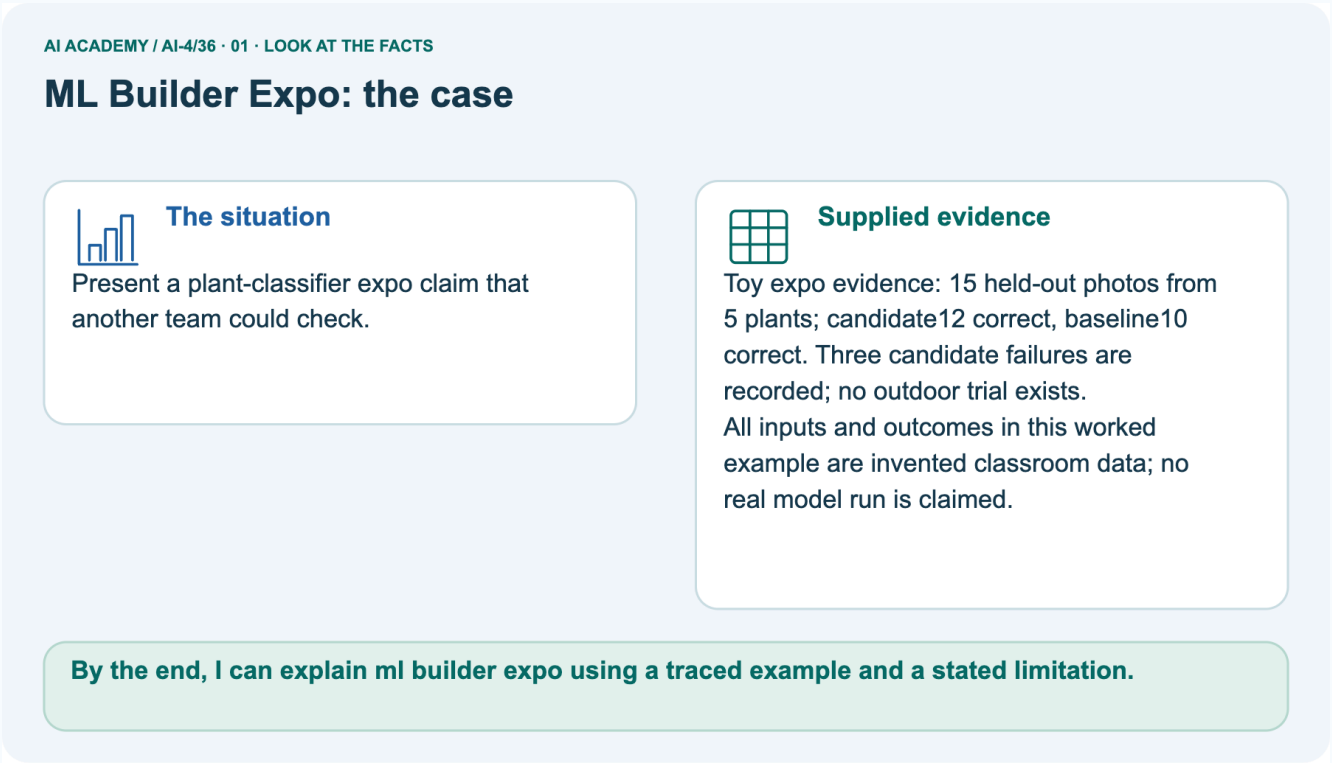
▼ Reference · The six-picture journey

REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the current question.

01. ML Builder Expo: the case



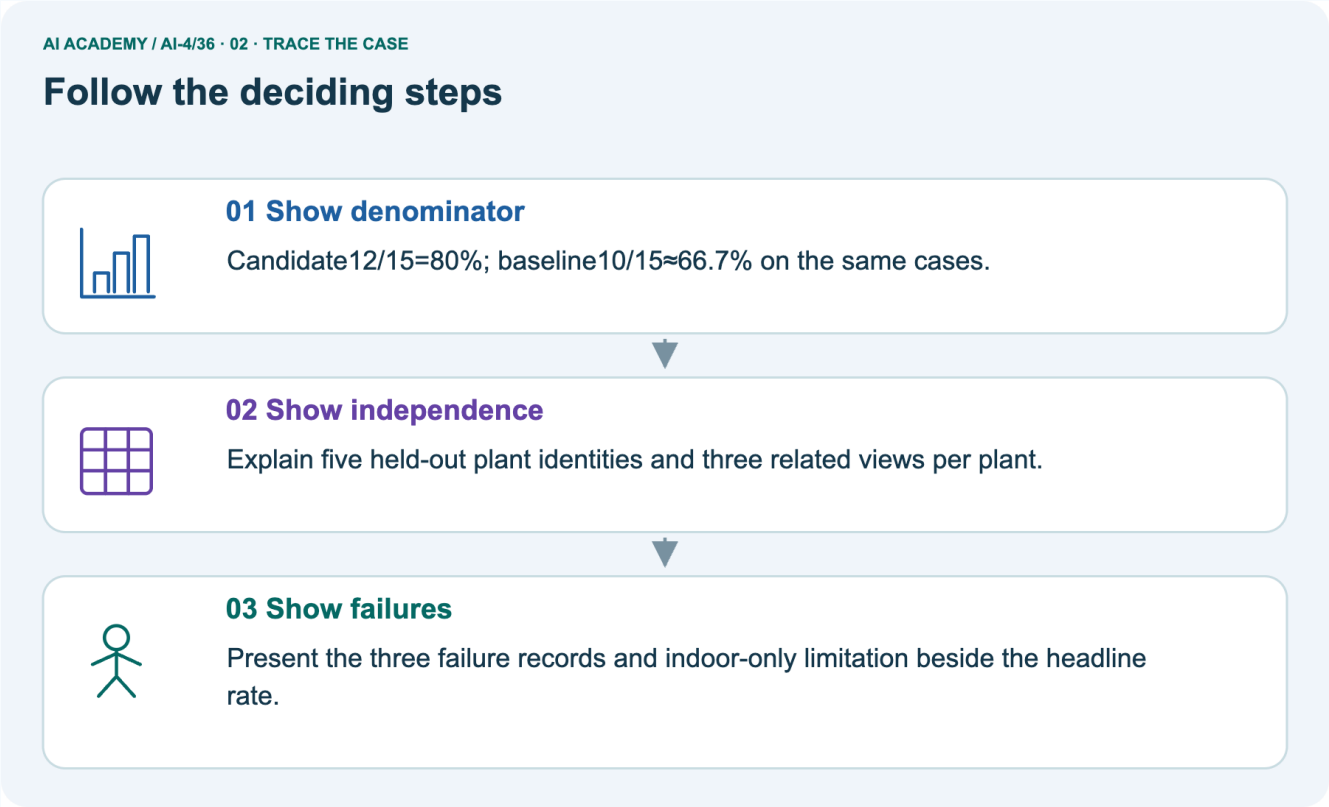
Follow the labelled picture with the matching explanation. Open the vector diagram

Present a plant-classifier expo claim that another team could check. Supplied facts: Toy expo **evidence**: 15 held-out photos from 5 plants; candidate12 correct, baseline10 correct.

Three candidate failures are recorded; no outdoor trial exists. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

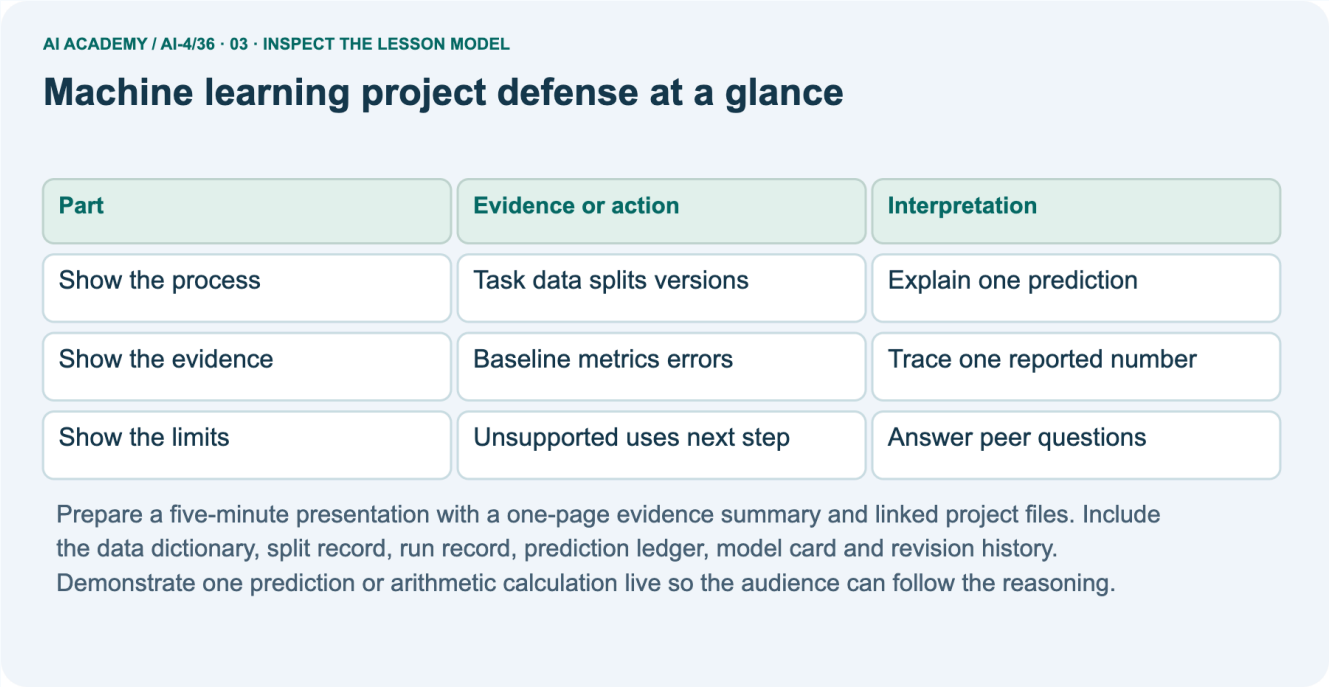


Follow the labelled picture with the matching explanation. Open the vector diagram

Show denominator: Candidate12/15=**80%**; baseline10/15≈66.7% on the same cases. → Show independence: Explain five held-out plant identities and three related views per plant. → Show failures: Present the three failure records and indoor-only **limitation** beside the headline rate.

Point to each numbered stage and explain which recorded **input** it uses.

03. Machine learning project defense at a glance



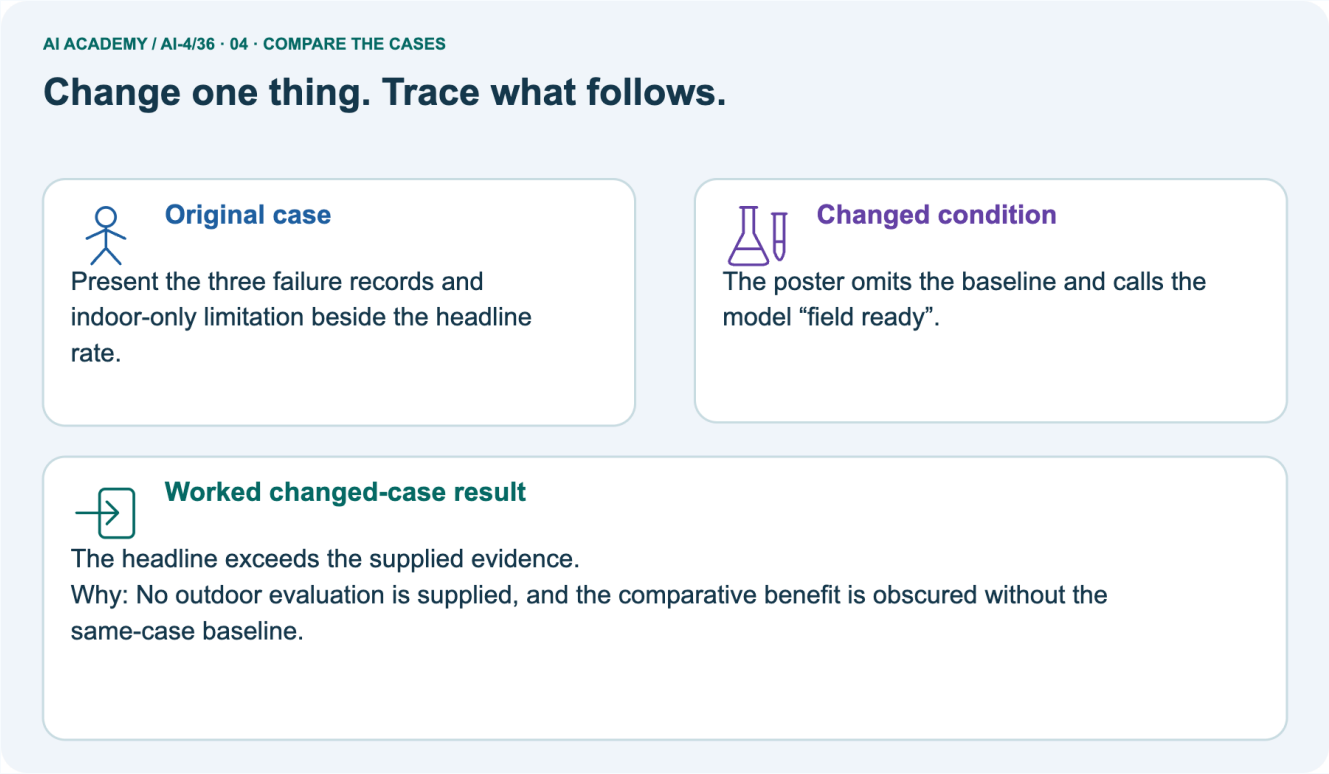
Follow the labelled picture with the matching explanation. Open the vector diagram

Machine learning project **defense** at a glance. Show the **process** | Task data splits versions | Explain one **prediction**. Show the **evidence** | Baseline metrics errors | Trace one reported number.

Show the limits | Unsupported uses next step | Answer peer questions

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



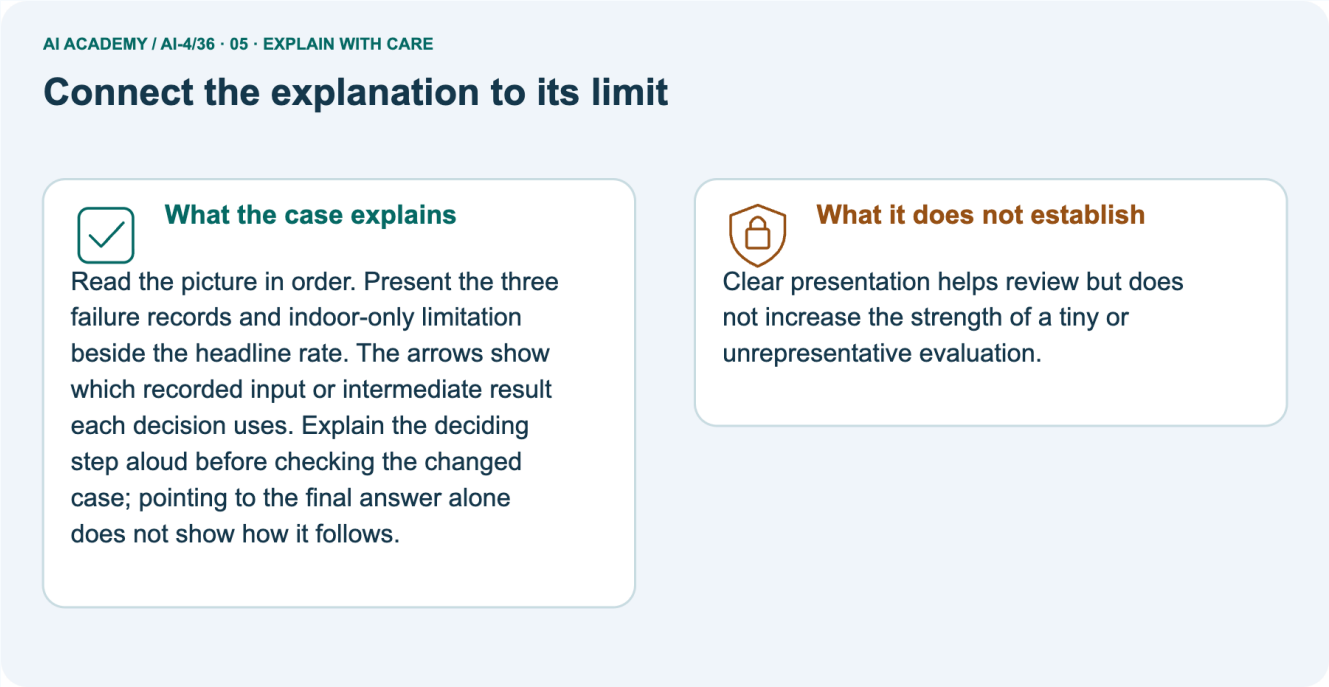
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Present the three failure records and indoor-only **limitation** beside the headline rate. Changed condition: The poster omits the baseline and calls the model “field ready”.

Expected result: The headline exceeds the supplied **evidence**. Reason: No outdoor evaluation is supplied, and the comparative benefit is obscured without the same-case baseline.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

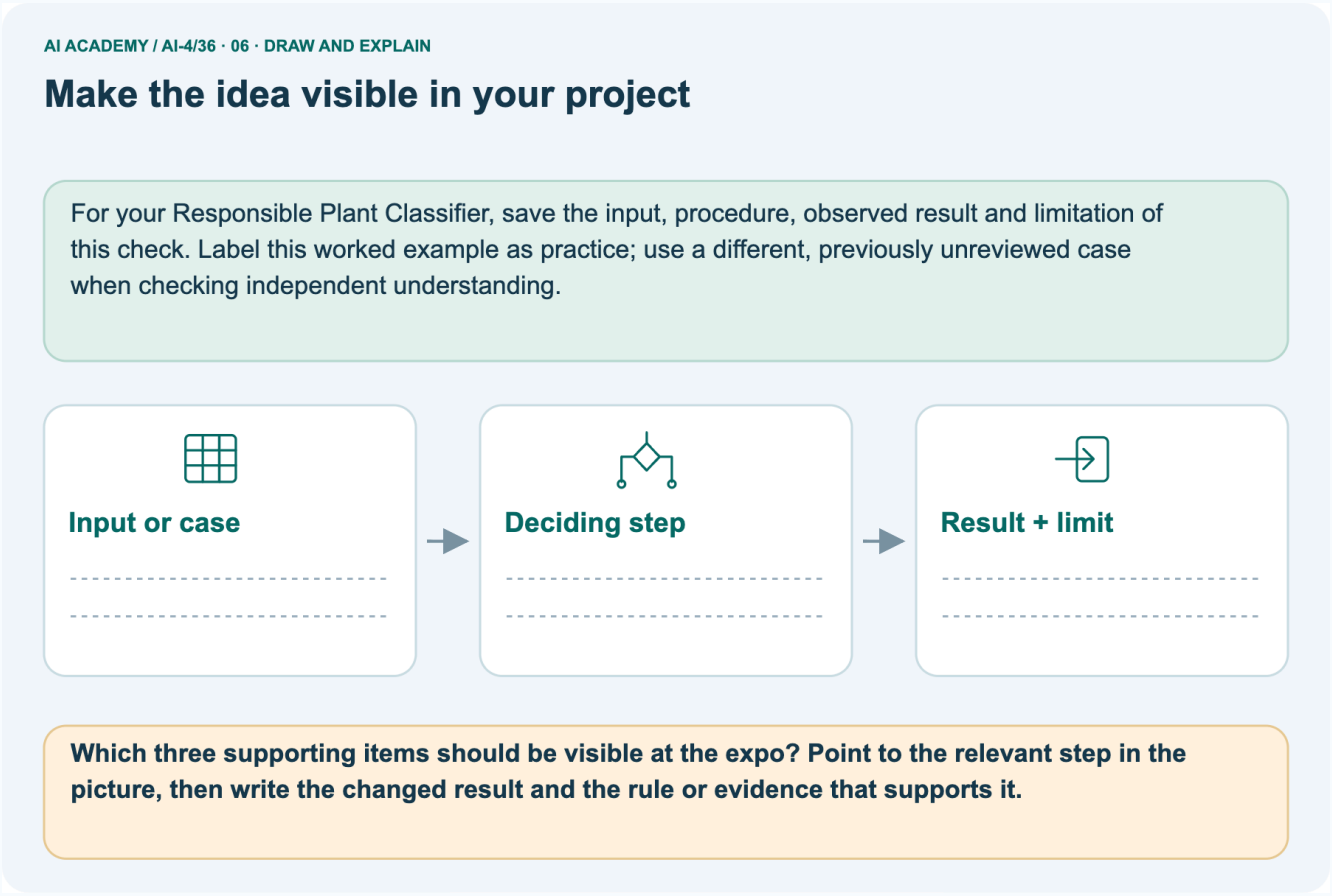
Read the picture in order. Present the three failure records and indoor-only **limitation** beside the headline rate. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Clear presentation helps review but does not increase the strength of a tiny or unrepresentative evaluation.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and **limitation** of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which three supporting items should be visible at the expo? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. ML Builder Expo: the case

02. Follow the deciding steps

03. Machine learning project defense at a glance

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.